

“I am going to give what I will call an elementary demonstration. But elementary does not mean easy to understand. Elementary means that very little is required to know ahead of time in order to understand it, except to have an infinite amount of intelligence.”

— Richard P. Feynman

MAGIC SQUARES UNLOCKED: From Puzzles to Patterns

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MAGIC SQUARES UNLOCKED:

From Puzzles to Patterns

EXCERPT:

From the ancient Lo Shu square of China to Dürer's Renaissance engraving, magic squares have captivated minds for centuries.

Beyond the well-known magic sum, these grids hide deeper wonders like the magic product and magic pairwise product, revealing unexpected mathematical patterns.

This talk takes you from building simple magic squares to exploring their hidden structures through stories, puzzles, and hands-on activities.

*Something “**ELEMENTARY**” can still hold endless magic,
if you look closely enough.*

What is a Magic Square?

A square grid of distinct numbers where every row, column, and both diagonals add up to the same number.

That number is called the '**Magic Constant/Number/Sum**'.

We'll now perform the highest form of

RESEARCH: PLAY ...

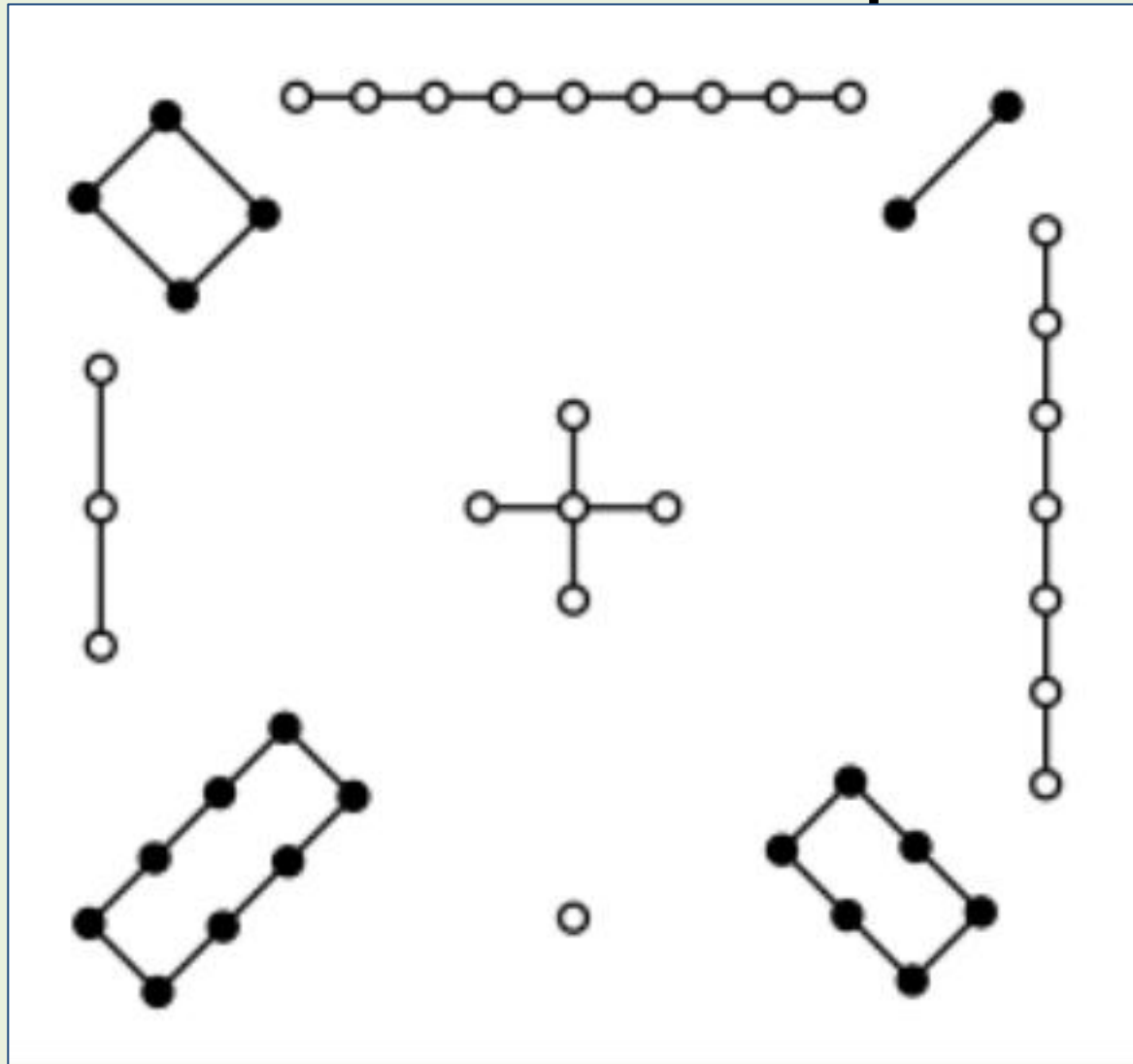
https://www.transum.org/software/SW/magic_square/magic_square.asp

2	7	6	→15	
9	5	1	→15	
4	3	8	→15	
↙15	↓15	↓15	↓15	↘15

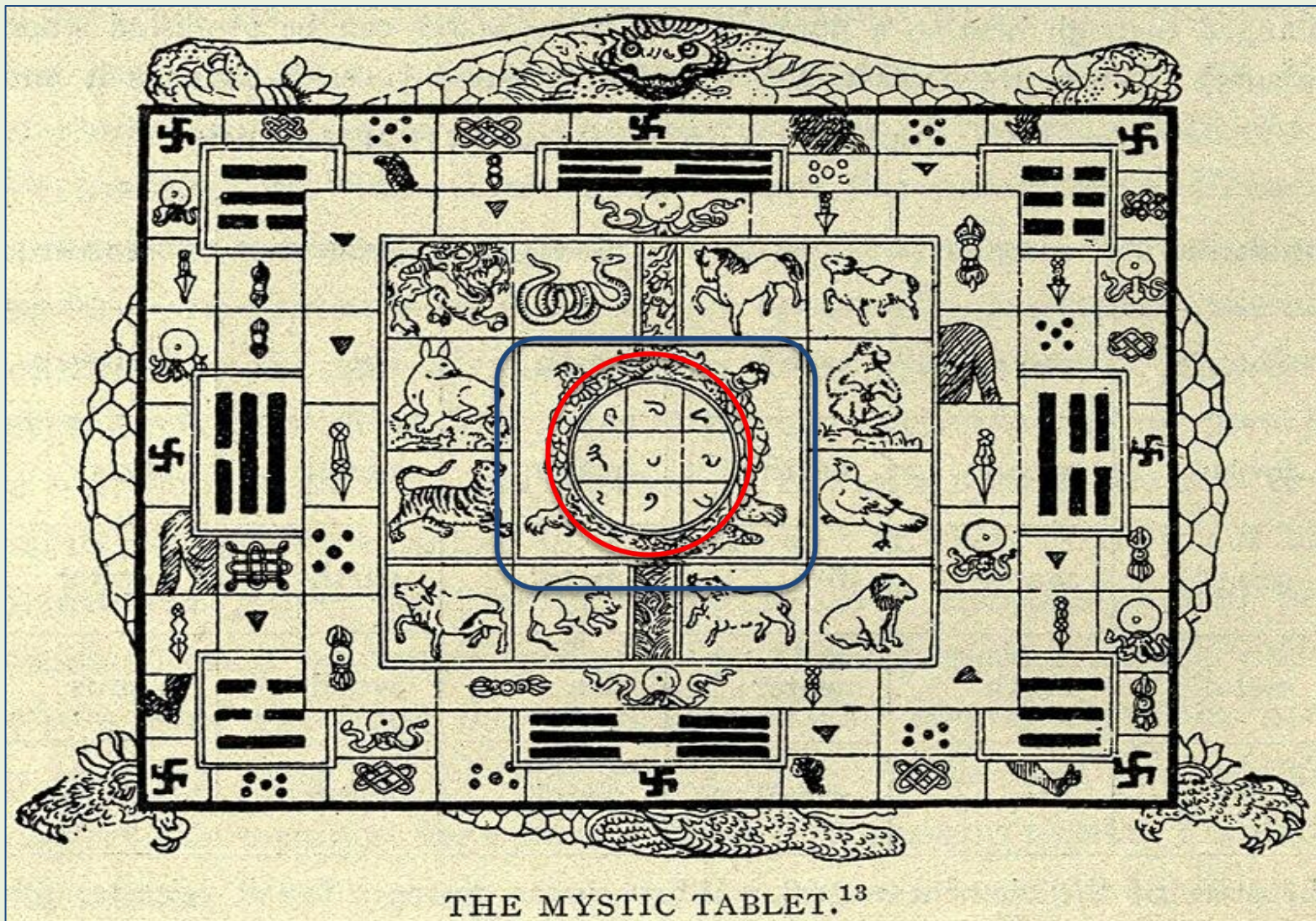
Magic Squares in Art & Culture

- **The Turtle, the Myth, the Legend:**
Lo Shu's Square (China)
- **The most perfect magic square of order 4:**
Parshvanatha Jain temple, Khajuraho (Madhya Pradesh, India)
- **Art and Magic square:**
Melencolia I (Painted by theorist of the German Renaissance:
Albrecht Dürer, 1514)
- **Perfume and Magic square:**
The oldest dateable fourth order magic square in the world
(an encyclopaedic work written by Varahamihira around 587
CE called Brhat Samhita.)

The Lo Shu 3×3 square



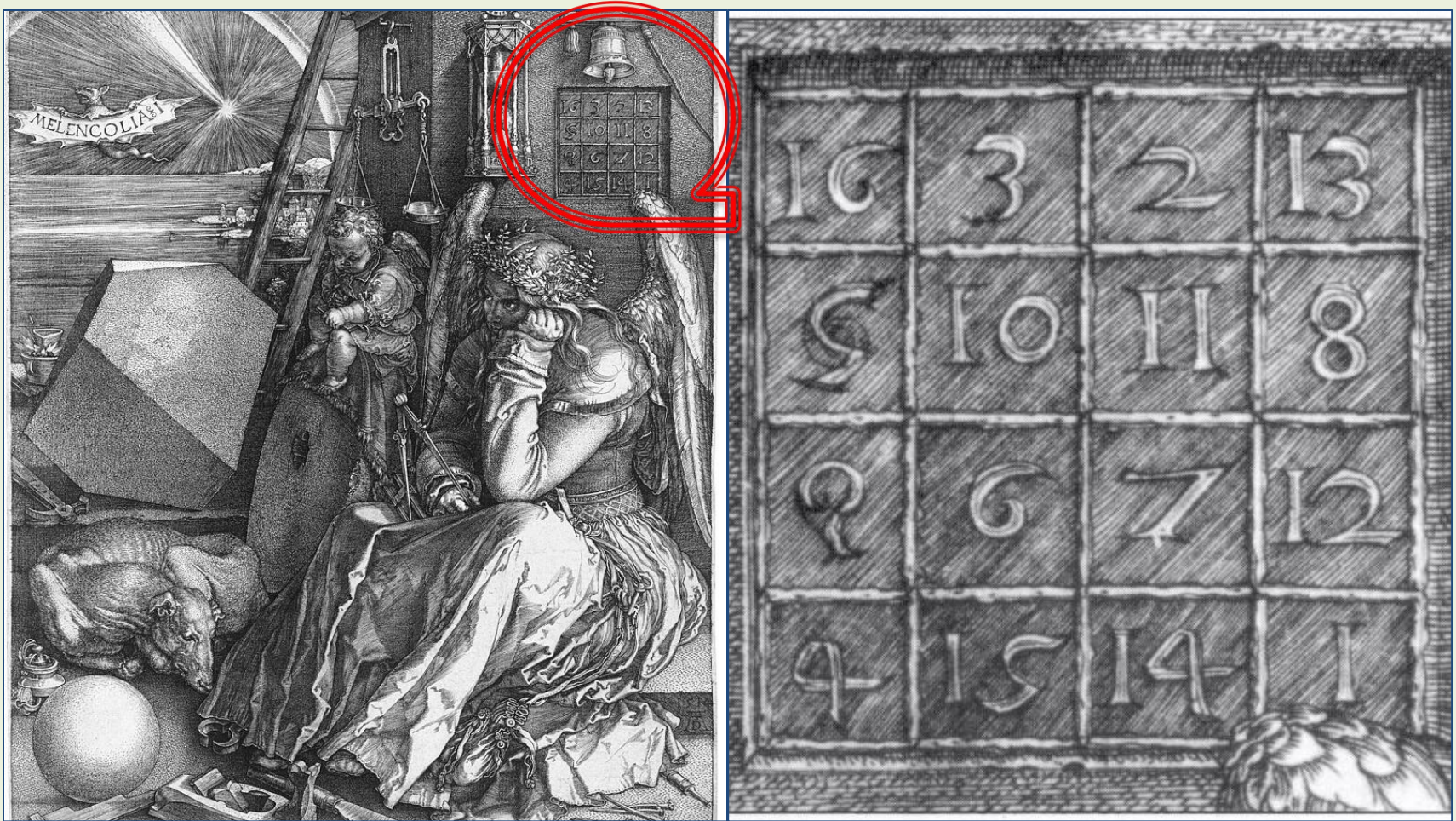
The Lo Shu 3x3 square on TURTLE-Shell



Parshvanatha Jain temple, Khajuraho (Madhya Pradesh, India)



Melencolia I (Albrecht Dürer, 1514)



Perfume and Math

This oldest dateable fourth order magic square in the world is found in an encyclopaedic work written by Varahamihira around 587 CE called Brihat Samhita.

- ❖ The magic square is constructed for the purpose of making perfumes using 4 substances selected from 16 different substances.

2	3	5	8
5	8	2	3
4	1	7	6
7	6	4	1

- Each cell of the square represents a particular ingredient.
- The number in the cell represents the proportion of the associated ingredient. So, the mixture of any four combination of ingredients along the columns, rows, diagonals, and so on, gives the total volume of the mixture to be 18.

In the square of Varahamihira the numbers 1 to 8 appear twice. So, It's a trivial magic square, and also a pan-diagonal magic square.

Some Questions to begin with...

- What happens to the Magic Sum of an order- n or $n \times n$ magic square if:
 - Each number of a magic square is increased by: ' k '
 - Each number of a magic square is multiplied by: ' k '
 - Each number ' m ' of a magic square is replaced by: ' $(n^2 + 1 - m)$ ' ?

*No need to worry if we can't answer immediately.
We'll start with $n = 3$. And soon we'll FIND SMARTER
WAYS for all order- n Magic Squares.*

It's Puzzle Time: LET, SET,GO...

- Now we'll EXPLORE the following MAGIC SQUARE related puzzles:

https://www.transum.org/software/SW/magic_square/magic_square.asp

- We'll MAKE WILD GUESSES, MAKE MISTAKES.
- We'll ADJUST our TECHNIQUES and REPEAT.

*This process is sometimes called:
LEARNING BY TRIAL AND ERROR.*

GOT ANY PATTERNS FROM PUZZLE?!

- For a classic order-3 or 3×3 magic square:
Magic constant/number/sum, $M = 15$ and central number = 5.
>> How to prove these?
- There is a formula for **Magic constant/number/sum** of a Classic order- n or $n \times n$ Magic Square: **$M = [n(n^2 + 1)]/2$**
So, for a 3×3 magic square where $n = 3$: $M = 3(3^2 + 1)/2 = 15$
>> How to prove the formula?
>> Will it work for $n = 2$??

Thinking Like a Mathematician:

CAN YOU FIND the 'Magic constant/number/sum'

and 'relation between Magic Sum and the central number' for any 3×3 magic square which are not classic (means the number used to create that magic square may be other than numbers from 1 to 9)?

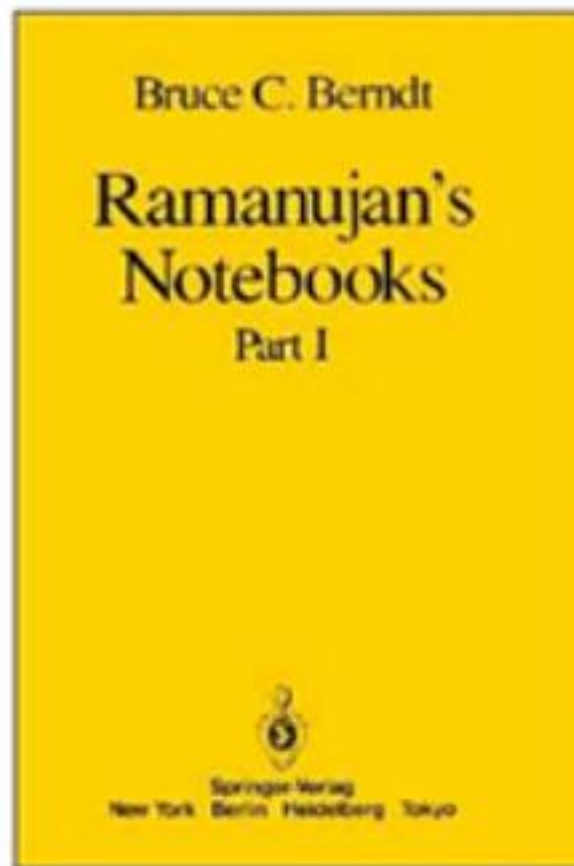
Let THE MAN (WHO KNEW ∞) Cook

Berndt 1985

B3-06

Ramanujan's work on magic squares is presented, in some detail, in Chapter 1 (pp. 16–24) of *Ramanujan's Notebooks, Part I*, by Bruce C. Berndt (Springer 1985)

"The origin of Chapter 1 probably is found in Ramanujan's early school days and is therefore much earlier than the remainder of the notebooks."

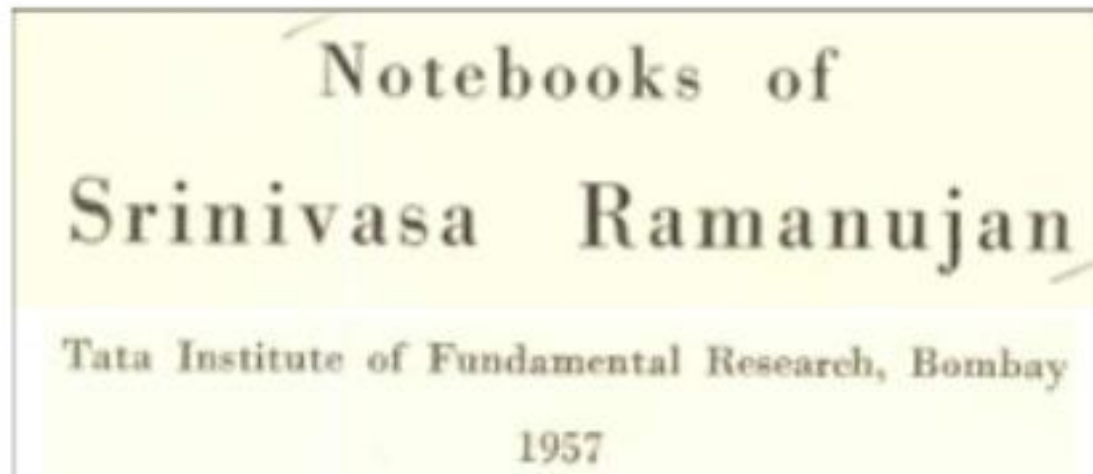


Let THE MAN (WHO KNEW ∞) Cook

Tata Institute 1957

B3-07

Ramanujan's work on magic squares was also presented,
photographed from its original form, in
Notebooks of Srinivasa Ramanujan,
Volume I, Notebook 1, and Volume II, Notebook 2,
pub. Tata Institute of Fundamental Research, Bombay, 1957.



Let THE MAN (WHO KNEW ∞) Cook

In Berndt 1985, Corollary 1, p. 17, we find:

In a 3×3 magic square,
the elements in the
middle row, middle column,
and each [main] diagonal
are in arithmetic progression.

Let THE MAN (WHO KNEW ∞) Cook

And so we have the general form for a 3×3 magic matrix

$$R_3 = \begin{pmatrix} h+u & h-u+v & h-v \\ h-u-v & \boxed{h} & h+u+v \\ h+v & h+u-v & h-u \end{pmatrix} = h\mathbf{E}_3 + u\mathbf{U}_3 + v\mathbf{V}_3$$

$$= h \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} + u \begin{pmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix} + v \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}.$$

Hands-on Puzzle & Reflection

Remember the following questions we were discussing earlier regarding MAGIC SQUARE?

“What happens to the Magic sum of an order- n or $n \times n$ magic square if:

- Each number of a magic square is increased by ‘ k ’
- Each number of a magic square is multiplied by ‘ k ’
- Each number ‘ m ’ of a magic square is replaced by: ‘ $(n^2 + 1 - m)$ ’ ?”

Now, we'll TACKLE them LIKE A BOSS!!!

PRO-PROBLEM SOLVER MOVE



PRO-PROBLEM SOLVER MOVE

What happens to the Magic sum of an order- n or $n \times n$ magic square if:

➤ Each number of a magic square is increased by 'k'

For, $n = 3$ we get a order-3 or 3×3 Magic Square:

Let, the Central Number of:

a 3×3 Magic square = h and Magic Sum = M .

Here, we can easily prove: $M = 3h$ (TRY THIS AT HOME)

Now, Each number of a magic square is increased by 'k'

So, let, the Central Number of:

the new 3×3 Magic square = h' and new Magic Sum = M' .

Hence, $M' = 3h'$ and $h' = h + k$,

Therefore, $M' = 3(h+k) = 3h + 3k = M + 3k$.

PRO-PROBLEM SOLVER MOVE

What happens to the Magic sum of an order- n or $n \times n$ magic square if:

➤ Each number of a magic square is multiplied by 'k'

For, $n = 3$ we get a order-3 or 3×3 Magic Square:

Let, the Central Number of:

a 3×3 Magic square = h and Magic Sum = M .

Here, we can easily prove: $M = 3h$ (TRY THIS AT HOME)

Now, Each number of a magic square is multiplied by 'k'

So, let, the Central Number of:

the new 3×3 Magic square = h' and new Magic Sum = M' .

Hence, $M' = 3h'$ and $h' = h \times k = hk$,

Therefore, $M' = 3(hk) = (3h)k = Mk$.

PRO-PROBLEM SOLVER MOVE

What happens to the Magic sum of an order- n or $n \times n$ magic square if:

➤ Each number ' m ' of a magic square is replaced by: ' $(n^2 + 1 - m)$ '?

For, $n = 3$ we get a order-3 or 3×3 Magic Square:

Let, the Central Number of:

a 3×3 Magic square = h and Magic Sum = M .

Here, we can easily prove: $M = 3h$ (TRY THIS AT HOME)

Now, Each number ' m ' of a magic square is replaced by:

' $(n^2 + 1 - m) = (3^2 + 1 - m) = (10 - m)$ '

So, let, the Central Number of:

the new 3×3 Magic square = h' and new Magic Sum = M' .

Hence, $M' = 3h'$ and $h' = 10 - h$,

Therefore, $M' = 3(10 - h) = 3 \times 10 - 3h = 30 - M$.

PRO-PROBLEM SOVLER MOVE

What happens to the Magic sum of an order- n or $n \times n$ magic square if:

- Each number of a magic square is increased by 'k'
- Each number of a magic square is multiplied by 'k'
- Each number 'm' of a magic square is replaced by: $(n^2 + 1 - m)$?"

Now, the main concern is whether the patterns and proofs found for $n = 3$ are true for all integer value of $n > 3$.

WHAT, WHY AND HOW DO YOU THINK?

IT'S CONJECTURE TIME! 2B OR NOT 2B?

We've used the fact that: "If the Central Number of:
a 3×3 Magic square = h and Magic Sum = M .

Then, $M = 3h$ "

Now, is it true for all integer value of $n > 3$??

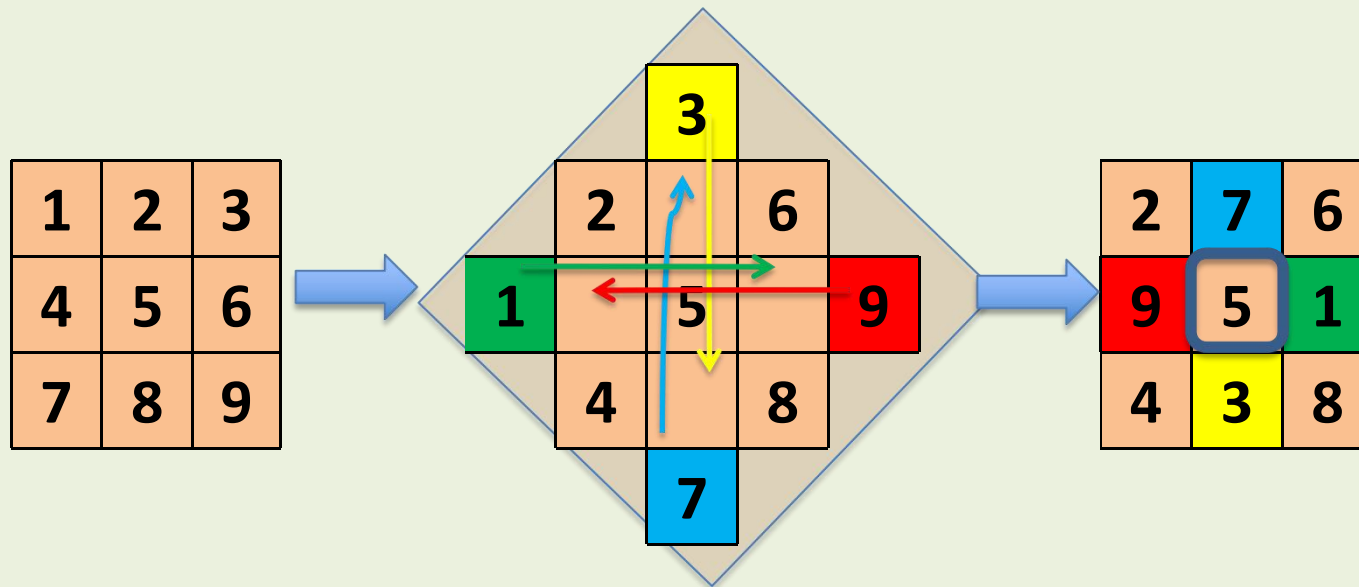
That, means can we claim this: "If the Central Number of:
a $n \times n$ Magic square = h and Magic Sum = M .

Then, $M = nh$ "

*Obviously, when n is an even number like 4,6,8...
there are no central number in a $n \times n$ Magic square.
So, let's conjecture it only for odd values of $n = 3,5,7,...$
YET THIS IS TOO GOOD TO BE TRUE. SO, LET'S VERIFY...*

Building Magic Squares for odd orders: Pyramid/Diamond Method (Cardano-Bachet)

Building a Classic Order-3 or 3×3 Magic Square



For, $n = 3$ we get a order-3 or 3×3 Magic Square:
The Central Number = $h = 5$ and Magic Sum = $M = 15$.
Then, $3h = 3 \times 5 = 15$. So, $M = 3h$.

Building Magic Squares for odd orders: Pyramid/Diamond Method (Cardano-Bachet)

Building a Classic order-5 or 5×5 Magic Square

*“TETRIS”
LOOKS TO ME*

			5				
		4		10			
	3		9		15		
2		8		14		20	
1		7		13		19	25
6		12		18		24	
	11		17		23		
		16		22			
			21				

For, $n =$
Square:



3	16	9	22	15
20	8	21	14	2
7	25	13	1	19
24	12	5	18	6
11	4	17	10	23

PROVED???
NOT SO FAST!!!

For, $n = 5$ we get a order-5 or 5×5 Magic Square: The Central Number = $h = 13$ and Magic Sum = $M = 11+12+13+14+15 = 65$
Then, $5h = 5 \times 13 = 65$. So, $M = 5h$.

Building Magic Squares for odd orders: Siamese or De la Loubère's Method

The Siamese method, or De la Loubère method, is a simple method to construct any size of n -odd magic squares (i.e. number squares in which the sums of all rows, columns and diagonals are identical). The method was brought to France in 1688 by the French mathematician and diplomat Simon de la Loubère, as he was returning from his 1687 embassy to the kingdom of Siam.

The Siamese Method or De la Loubère's Method makes the creation of magic squares straightforward.

Building Magic Squares for odd orders: Siamese or De la Loubère's Method

"Mr. Vincent, whom I have so often mentioned in my Relations, seeing me one day in the ship, during our return, studiously to range the Magical Squares after the method of Bachet, informed me that the Indians of Suratte ranged them with much more facility, and taught me their method for the unequal squares only, having, he said, forgot that of the equal"

— *Simon de la Loubère, A new historical relation of the kingdom of Siam.*

Building Magic Squares for odd orders: Siamese or De la Loubère's Method

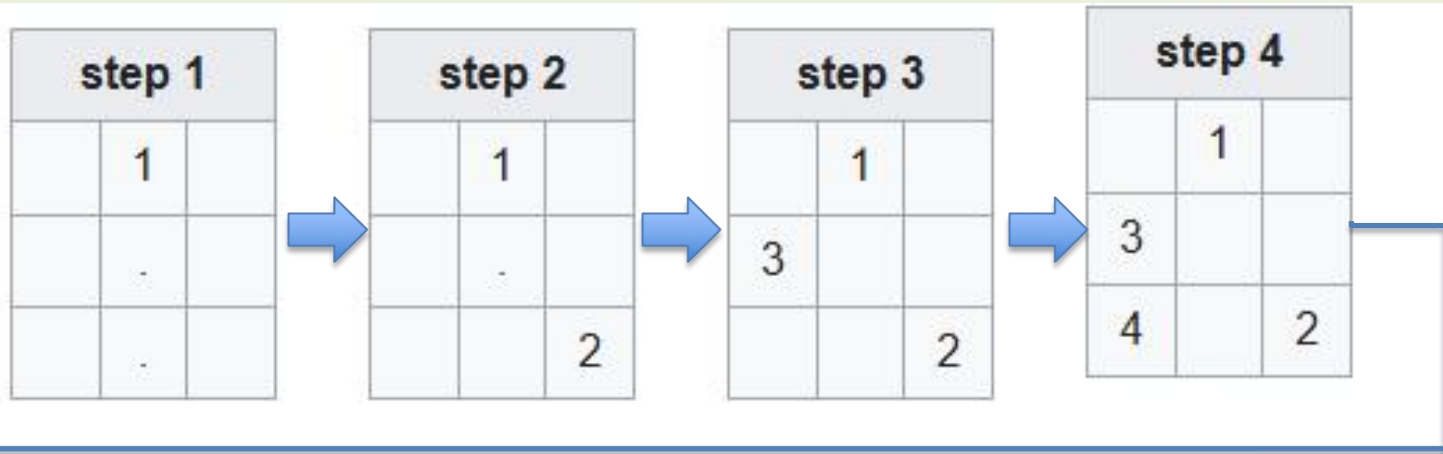
ALGORITHM:

1. Place 1 in the middle of the top row
2. Move up and right for the next number
3. Wrap around edges
4. If space is occupied, move down instead

Let's see how it works:

<https://en.wikipedia.org/wiki/File:SiameseMethod.gif>

Building Magic Squares for odd orders: Siamese or De la Loubère's Method



Building Magic Squares for odd orders: Siamese or De la Loubère's Method

Tome II. *of the Kingdom of SIAM.*

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towards the right: that is to say, that from the upper transverse they descend immediately to that below.

3. When they have placed a number in the last case of a transverse, the following is put in the first case of the transverse immediately superior, that is to say, that from the last upright, they return immediately to the first upright on the left.

4. In every other occurrence, after having placed a number, they place the following in the cases which follow diametrically or slantingly from the bottom to the top, and from the left to the right, until they come to one of the cases of the upper transverse, or of the last upright to the right.

5. When they find the way stopp'd by any case already filled with any number, then they take the case immediately under that which they have filled, and they continue it as before, diametrically from the bottom to the top, and from the left to the right.

These few Rules, easie to retain, are sufficient to range all the unequal Squares in general. An example renders them more intelligible.

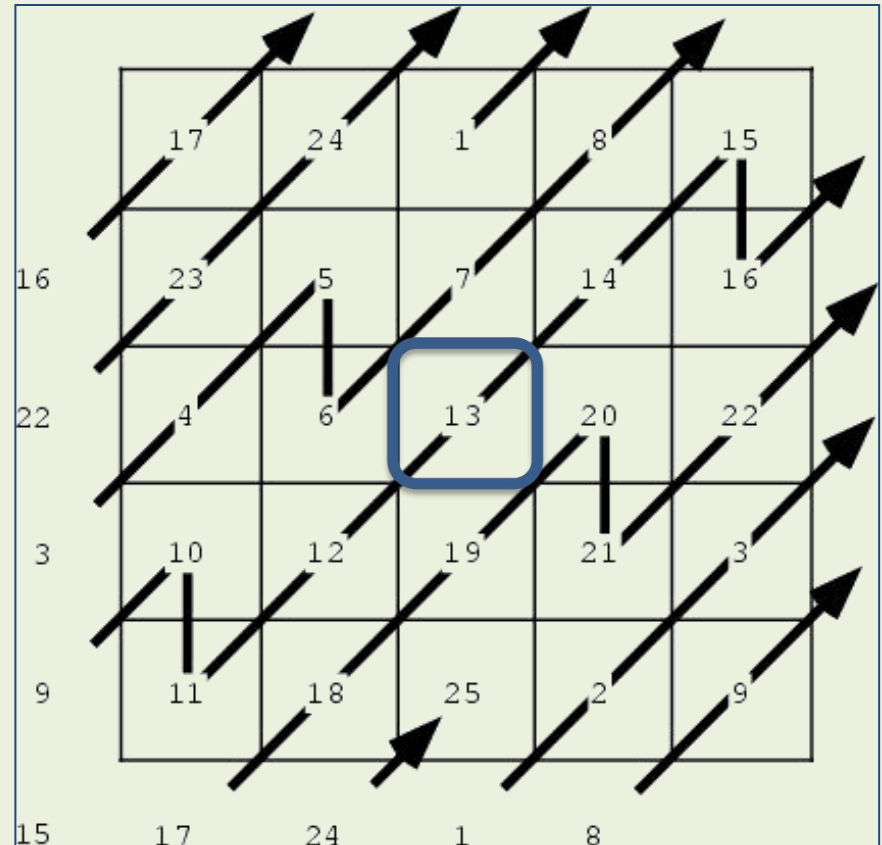
17	24	1	8	15
23	5	7	14	16
4	6	13	20	22
10	12	19	21	3
11	18	25	2	9

This square is essentially different from that of *Agrippa*; and the method of *Bachet* is not easily accommodated thereto; and on the contrary, the *Indian* method may easily give the squares of *Agrippa*, by changing it in something.

1. They place the unite in the Case, which is immediately under that of the Center, and they pursue it diametrically from top to bottom, and from the left to the right.

2. From the lowest case of an upright, they pass to the highest case of the upright which follows on the right; and from the last case of a Transverse they return to the left to the first case of the Transverse immediately inferior.

3. When the way is interrupted, they re-assume two cases underneath that which they filled; and if there remains no case underneath, or that there remains but one, the first case of the upright is thought to return in order after the last, as if it was indeed underneath the lowest.



PROVED NOW? CONVINCED?

TRUST AND BETRAYAL

We've solved 5×5 magic square in 2 different way and in both cases we get: **the central element = $h = 13$.**

*COOL PATTERN, RIGHT? BUT PATTERNS CAN BETRAY.
BEWARE of TRUSTING A CONJECTURE WITHOUT PROOF.*

13	11	15	16	10	⇒	65
17	5	19	18	6	⇒	65
14	22	1	8	20	⇒	65
9	24	7	21	4	⇒	65
12	3	23	2	25	⇒	65



65	65	65	65	65	65
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For, $n = 5$ we get this
order-5 or 5×5 Magic Square:
The Central Number = $h = 1$
and Magic Sum
 $= M = 12+24+1+18+10 = 65$
Then, $5h = 5 \times 1 = 5$.
So, $M \neq 5h$

ALGEBRA saves LIFE

4x4 MAGIC SQUARE COMPLETING

Can the following partial square be completed to obtain a magic square of order 4? You've to use 1,2,3,4,....., 16 to complete this magic square. Create the magic square. Also if its not possible to create a magic square in the given circumstances then show your logic in details.

4			6
10			8

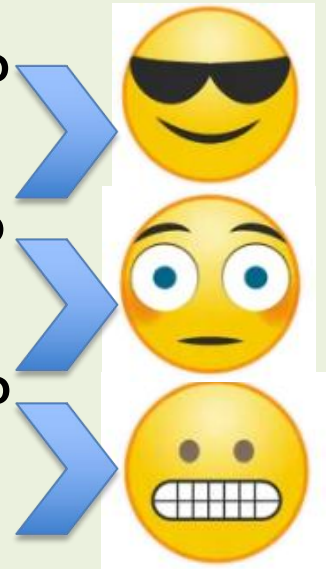
METACOGNITIVE REFLECTION:

- What Challenge have you Faced?
- What Strategies have you used or thought of using?
- What you've learned by solving this problem?

Questions that won't let you sleep?

Let's count the number of classic magic squares involving rotation and reflection.

- How many distinct 3×3 magic squares exist?
YOU CAN DO IT. TRY NOW.
- How many distinct 4×4 magic squares exist?
YOU CAN. DO IT. TRY THIS AT HOME.
- How many distinct 5×5 magic squares exist?
CAN YOU DO IT? DON'T TRY THIS AT HOME.



Thinking Like a Mathematician:
Will ALGEBRAIC FORMULATION &
GEOMETRIC SYMMETRY save the day?

Beyond Sums : Magic Product

8	1	6
3	5	7
4	9	2

8	1	6
3	5	7
4	9	2

- **Lo Shu Magic Square example:**

➤ Multiply the three numbers in each row together and add the three products:
 $(8 \times 1 \times 6) + (3 \times 5 \times 7) + (4 \times 9 \times 2)$
 $= 48 + 105 + 72 = 225$

➤ Multiply the three numbers in each column together and add the three products:
 $(8 \times 3 \times 4) + (1 \times 5 \times 9) + (6 \times 7 \times 2)$
 $= 96 + 45 + 84 = 225$

This '**225**' is called: '**Magic Product**'.

Beyond Sums : Magic Pairwise Products

8	1	6
3	5	7
4	9	2

8	1	6
3	5	7
4	9	2

- **Lo Shu example:**

- Take all pairwise products in each row and add them up:

$$(8 \times 1 + 1 \times 6 + 6 \times 8) + (3 \times 5 + 5 \times 7 + 7 \times 3) + (4 \times 9 + 9 \times 2 + 2 \times 4) = 62 + 71 + 62 = 195$$

- Take all pairwise products in each column and add them up:

$$(8 \times 3 + 3 \times 4 + 4 \times 8) + (1 \times 5 + 5 \times 9 + 9 \times 1) + (6 \times 7 + 7 \times 2 + 2 \times 6) = 68 + 59 + 68 = 195$$

This '**195**' is called: '**Magic Pairwise Product**'.

Hands-on Puzzle & Reflection

- CAN YOU FIND the relation between magic sum and magic pairwise sum of a 3×3 magic square?
- CAN YOU FIND two 3×3 magic squares with the same magic sum but two different magic products?
- CAN YOU FIND two 3×3 magic squares with the same magic product but two different magic sums?
- CAN YOU FIND two 3×3 magic squares with the same magic sum and magic product but two different magic pairwise products?

Final Challenge

- **Challenge:**

An Elementary puzzle that needs infinite intelligence:
"Can you create a Magic square using only square numbers?"

- **Thinking Like a Mathematician:**

*First, try to answer if it's possible to create a 3×3 magic square using these 9 squares numbers:
 $1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2, 8^2, 9^2$?*

IF you think "YES" THEN show one such example.

IF you think "NO" THEN prove that.

(Thanks+Resources) for Thinking

- https://en.wikipedia.org/wiki/Magic_square
- https://www.transum.org/software/SW/magic_square/magic_square.asp
- <https://www.rigb.org/explore-science/explore/blog/fascination-magic-squares>
- https://en.wikipedia.org/wiki/Most-perfect_magic_square
- <https://en.wikipedia.org/wiki/File:SiameseMethod.gif>
- <https://www.math.mcgill.ca/styan/Beamer3-18jan12-opt.pdf>
- NRICH: Magic sums and products:
<https://nrich.maths.org/articles/magic-sums-and-products>
- Martin Gardner's work on recreational math:
<http://www.multimagie.com/English/SquaresOfSquares.htm>